TECHNICAL & FINANCIAL PROPOSAL

RemitHome - API Interfacing (PULL Model)

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1 Proposal

1.1 Request for Proposal

1.1.1 The request for proposal was received from JSBL in 18 June 2015.

1.2 Scope of Work

1.2.1 Two remitting entities ('EPS' and 'SpeedRemit') will be covered under the covered scope of PULL API model and for this no additional fees will be charged.

Background
Refer RH-JSBL- Technical Proposal dated 20 June 2012, clause # 1.6.2 as per which , we are going to provide JSBL API interfacing functionality with two entities (EPS and Speed Remit) only under covered original scope.

Integration with Remitting Entities /Agents (MTOs) through APIs:
Following remitting entities APIs have been shared and reviewed
- o EPS
- o Speed Remit

Note: Further, any additional development requested by JS will be cover under separate scope of work and charge accordingly)

1.2.2 The scope, time and fee for any additional remitting entities which JSBL wants to integrate with 'RemitHome' application will be determined separately.

1.2.3 The PULL API based on a web-service interface to facilitate JSBL for connecting with remitting entities through web service integration with 'RemitHome' and get transactions for further processing.

1.2.4 'RemitHome' Web Service API (PULL Model) supports following transaction types
- Account transactions (includes both JSBL and other bank transactions)
- Cash over the counter payments

1.2.5 'RemitHome' web services integration frame work will enable JSBL to receive real time instant remittances from the different remitting entities system (On payment from anywhere concept).

1.2.6 Under this model transaction will not reside in 'RemitHome' Database. Upon request of Beneficiary, the application will pull the transaction from remitting entity system through beneficiary provided UNR. Once Transaction is paid, acknowledgement will be sent to remitting entity system through API on real time bases.

1.3 Application Coverage

1.3.1 JSBL want 'RemitHome' to integrate with its exchange Companies through APIs and process the data on real time bases. All remitting entities will be using Web services to expose their methods and require some level of authentication. The web methods are using different t modes of data exchange like XML, String, Object, etc

1.3.2 New module will be introduced in existing 'RemitHome' application for setting up any new remitting entity interfacing with Web Service API.
These Includes:
- Entry Form to setup Remitting Entity for allowing connectivity with credential information.
- Entry Form for Security Settings.
- Inquiry screen to monitor API transactions.

1.3.3 API Interfaces Available in Application

Usually three set of messages required to complete single transaction (but it may differ from entity to entity).
a- Transaction Inquiry
b- Confirm Payment
c- Cancel Transaction

1.3.4 Transaction Processing

There are two types of processing modes,

1.3.3.1 COTC payment through API:
COTC payment interface will be used as first interaction point at branch level where beneficiary comes to collect remittance. It is used to capture details related to transaction and beneficiary.

For all of those transactions which are related to remitting entities having API interfacing with Bank, drop down is used to select transaction originating remitting entity and forward request to Remit Home API for further action.

1.3.3.2 *Account Credit transaction processing through PULL API:*
PULL API will be performing the below steps to download the Account Credit Transaction. A service will be scheduled to generate asynchronous entity wise request(s) to download Own Account Transactions.

Note: Like COTC Pull API, there is no request initiator in Account Pull API, so a scheduled service will be initiating the request for downloading the transactions

API will make call to remitting entity API for downloading the transactions which are outstanding or available for bank.

While downloading the transaction, API will retrieve the data from returned data set on the basis of PRI Field & Response mapping setup,

API will use same validation rules of 'RemitHome' File Downloading Process to download the transaction. Following validations rules would be applied:

- Special Characters validation in all fields
- Account Length Validation
- Unique Reference validation for Null Values
- Remitter & Beneficiary Name validation for Null values

Another service will also be scheduled for marking the status (as OK or Rejected) of recently downloaded transactions into remitting entity API

Note: Some Remitting Entities do not provide the method of marking the Transaction Status. If no status marking method is available then it has to be done manually.]

Also at later time this service will be responsible for marking the updated/modified transaction status (for e.g. PAID) into a remitting entity API.

It will be the responsibility of Tie-ups, that they will not send the data again once they have send it at JSBL.

For both COTC and account credit transactions, Core banking integration will be required which will be done through web services exposed by JSBL.

1.4 Centegy Offer

1.4.1 Our offer is the licensing of 'RemitHome-Web based API (PULL Model)' application to one legal entity (which is JSBL operating in Pakistan) as per the agreed scope of work (refer section 1.2).

1.4.2 Centegy technologies grants to the client a perpetual, personal, non-transferable and non-exclusive right and license to use the Software, at the location(s) identified herein. The CLIENT represents warrants and agrees that the Software will be used only at such location(s) and only for the benefit of the client.

1.4.3 Centegy Technologies will not be responding to any HW (Hardware) related queries as part of its response (i.e. not providing hardware specification and cost). Our response is limited to its Home Remittance 'RemitHome' application software solution.

1.4.4 Provision is available to embed End to End SSL encryption with the application
Note: The SSL certificate acquisition, configuration and maintenance will be client responsibility.

1.4.5 The business partners of the Bank shall be entitled to access only the software from the centralized location of the bank in order to perform their business operations.

Note: The Bank should have to acquire the application license for their Business Partners. The licensing cost and terms/condition are defined in the financial section with this proposal.

1.4.6 We are not responsible for providing any training or implementation support to such business partners. If required, separate scope, time and fee will be ascertained upon receiving request.

1.4.7 The application shall be installed and implemented at Bank's site in Karachi, Pakistan and may be relocated by the Bank at any other of its branch after prior written intimation to us as provided in the Agreement. However, we are not responsible for the re-installation, configuration, training or implementation at any new location.

1.5 Specific/Customized Integration

1.5.1 The Time and Fee to complete the scope covered under enhancements/customization will determined after analyzing scope like

- Integration with Bank Own Core Banking Application
- Integration with Bank AML Software
- Integration with Remitting Entities /Agents (MTOs) through APIs
- Integration with Bank SWITCH for IBFT
- Integration with Bank SMS solution

1.5.2 This scope of work includes review, development and user acceptance testing.

2 Financials

2.1 Fee Quotation

No.	Fee Components	Fee (Rs.)
01	Remitting Entities – Covered under Scope of Work (PULL API) Speed Remit and EPS Note: A maximum of 45 calendar days will be assigned for single remitting entity (from development, release deployment and UAT Sign-Off) after determination of scope and its acknowledgement received from client.	000/-
02.	Minimum Fee for the efforts required in PULL API Interfacing with Other Remitting Entities (Customized Development). This is the per Remitting Entity API Interfacing Fee which includes Review, Configuration, Development and UAT Note: A maximum of 45 calendar days will be assigned for single remitting entity (from development, release deployment and UAT Sign-Off) after determination of scope and its acknowledgement received from client.	550,000/- Per Remitting Entity
03.	Additional Fee for the efforts required in PULL API Interfacing with Remitting Entities (covered under scope or not) in case the UAT Sign-off not received within 45 days after determination of scope and its acknowledgement received from client.	25,000/- Per Day x No of Man Days

2.1.1 The fee is exclusive of all local taxes and would be adjustable subject to changes in the scope.

2.1.2 The fee of the assignment does not include the cost of Hardware, Operating System License fee, Database Engine License Fee, and Front-End Tool Software License.

2.1.3 As per CENTEGY's general policy, we do not provide source code; therefore, the fee mentioned above is without source code.

2.1.4 As per our standard terms and conditions we have offer free warranty period of six (6) calendar months that will be commenced from the time of application installation.

2.2 Other Fee's

2.2.1 If the Bank requires enhancement or customization or integration at any time, a 'BRD – Business Requirement Document" will be initiated by Bank. In response to this CENTEGY, shall provide 'Change Request – CRF' having scope and cost estimate and will commence work as soon as possible after the CRF has been accepted by the client. The development charges (per man day) will be determined on the basis of scope of work and man days efforts required for any review, enhancement, customization, integration, etc. activity.

2.2.2 Additional fee will be charged for the specific Integration with Other Software Applications like Core Banking interfacing, RE/MTOs (API Interfacing), SWITCH Interfacing, AML Interfacing, SMS Interfacing, etc. The review, customization, configuration, support fee will be charged as per the agreed chargeable rate (per man day) separately, depending on the nature of work and man days efforts involved.

2.2.3 The software maintenance services on expiry of warranty period, under a separate agreement will require to be signed. The terms and condition for the maintenance fee for this proposal will be the same as per main maintenance contract, submitted to client for finalization.

2.2.4 In addition, we will charge out of pocket expenses (for visits outside Karachi), if any, which includes traveling, boarding and lodging cost at actual plus the per diem cost of @ Rs. 2,000/- per person per day for our staff traveling during the period of this assignment and extended support period. The 'OOP' fee for any international visit will be determined separately (if required).

2.3 Payment Terms

No.	Schedule	Per
	RemitHome API (PULL Engine) License Fee	
01-A.	Upon Acceptance of Proposal	50%
01-B.	Upon Installation/Deployment	50%
	RemitHome (PULL Engine) API Customized Interfacing Fee	
02-A.	Upon Acceptance of CRF	50%
02-B.	Upon Completion of UAT	50%

3 Other Matters

3.1 Specific

3.1.1 The proposal is valid for acceptance until a period of Two (02) months from the proposal date.

3.1.2 We will be in a position to commence work on the assignment within four weeks after receipt of Purchase Order.

3.2 General

3.2.1 The provision of suitable working space, environment with access to computer systems as well as stationery and backup media for our project team for all aspects will be the responsibility of the client.

3.2.2 The client, during the execution of the assignment or for a period of one year following the end of the assignment, will not make or attempt to make an offer of employment to any of our employees, either those working on the assignment or otherwise.

3.2.3 We would recommend that the company appoint a liaison person for this assignment whom we can contact for information when necessary. An effective liaison will greatly speed up the completion of the assignment.

3.2.4 We do not make any warranties, expressed or implied, towards successful implementation of the project / software, as we understand that besides the software there are a number of other variables that will contribute towards an effective implementation. However as per tradition and practice our best efforts & resources will be deployed towards generation of desired results.

3.2.5 For multi-location deployment, setting up of Wide Area Network and connectivity will be the responsibility of the client. Likewise, database administration, replication, backup and optimization will also be handled by the client resources.